Survival Analysis in Healthcare

Kaplan-Meier, Hazard Ratios & Clinical Risk Stratification

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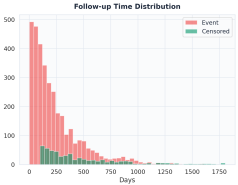
Agenda

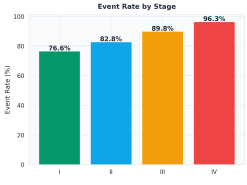
1. Clinical Context
2. Kaplan-Meier Analysis
3. Hazard Ratios
4. Predictive Modeling
5. Conclusions

Study Design

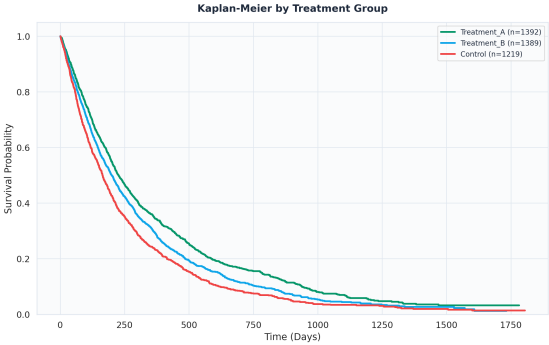
Dataset

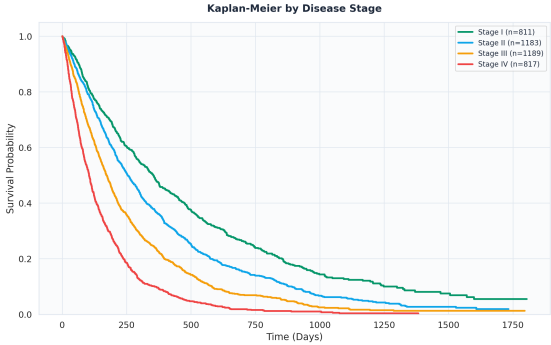
- **4,000 patients**, 14 clinical features
- 3 treatment arms + disease staging I–IV
- Event rate: 86.4% (14% censored)
- Follow-up: up to 5 years
- Covariates: age, BMI, smoking, biomarkers, comorbidities





Survival by Treatment & Stage

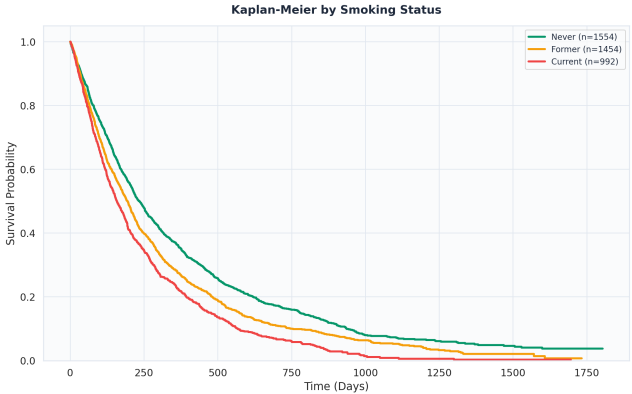




Key Findings

Treatment A shows a clear **survival benefit** over Control. Stage IV patients have dramatically worse outcomes (median survival <6 months vs >18 months for Stage I).

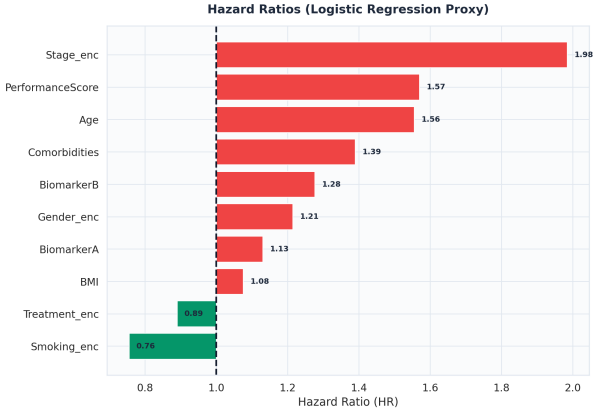
Survival by Smoking Status



Smoking Impact

Current smokers show significantly worse survival than never-smokers, with former smokers intermediate. This is an **independent risk factor** after adjusting for stage and treatment.

Risk Factor Analysis

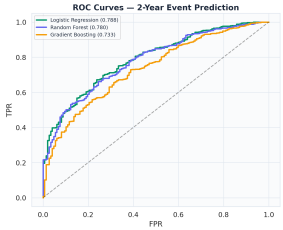


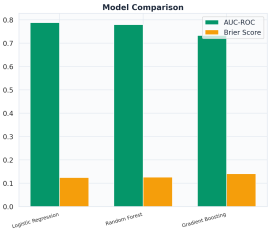
Interpretation

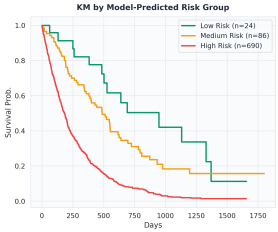
HR > 1: increased risk. HR < 1: protective.

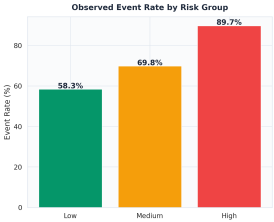
- **Stage**: strongest predictor (HR up to 3.3)
- **Performance score**: significant prognostic factor
- **Treatment**: protective effect (HR $<$ 1)
- **Smoking**: dose-response relationship

2-Year Event Prediction & Risk Stratification





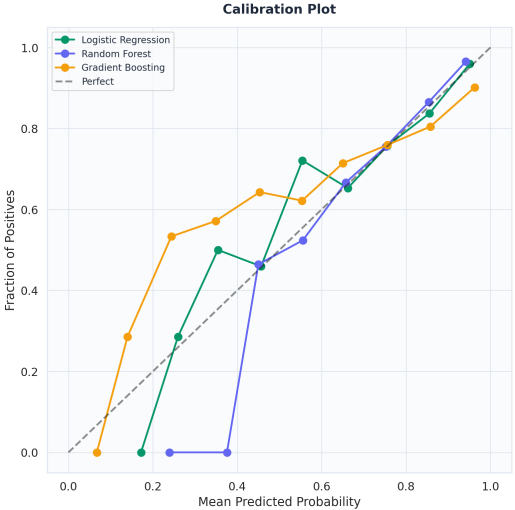




Clinical Utility

Model-predicted risk scores stratify patients into distinct prognostic groups. High-risk patients have 3x+ event rate vs low-risk — enabling **targeted intervention**.

Calibration



Well Calibrated Predictions

Summary & Future Work

Findings

- Treatment A: significant survival benefit
- Stage IV: 3x+ hazard vs Stage I
- Smoking: independent risk factor
- AUC 0.79 for 2-year prediction
- Risk stratification validated by KM

Future Work

1. Cox PH with lifelines library
2. Competing risks analysis
3. Time-varying covariates
4. External validation cohort
5. Nomogram for clinical use

Thank You

Questions & Discussion

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